

3 (**long**) Figure 1 shows a horizontal axis piston and cylinder arrangement. The cylinder contains argon, initially at 1 bar and 300 K (p_1, T_1) in an environment also at 1 bar. The cylinder diameter is 0.1 m and the mass of argon is 0.01 kg. Initially, the spring is unloaded and has a length of 1 m. It has a spring constant k of 200 Nm^{-1} . The piston may be assumed to be frictionless and there is no heat transfer, except as described in (b) below. You may assume that argon is a perfect gas.

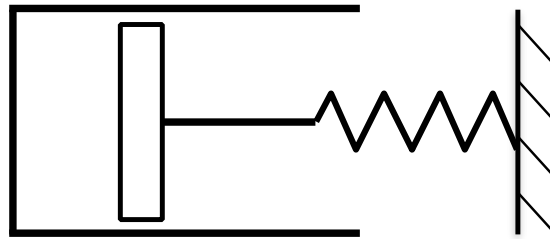


Fig. 1

- (a) What is the initial volume V_1 of the trapped gas? [6]
- (b) The gas is slowly heated until the spring length is 0.5 m. Calculate the final gas temperature. Also, calculate the work done by the gas and the heat transferred to the gas during the process. [12]
- (c) If, instead of process (b), the piston is displaced from its initial rest position by a small distance and then released, show that the subsequent oscillations of the piston will be as a spring-mass system with an effective spring constant of $(k + \gamma p_1 A^2 / V_1)$, where γ is the ratio of the gas specific heat capacities. It may be assumed that the expansion/compression oscillation process is isentropic. [12]